Mechanical waves and distant effects

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use travel-time reasoning while distinguishing waves from the motion of material.

SUCCESS CRITERIA - I can...

1. Identify the medium carrying the wave.
2. Use speed and distance to estimate travel time.
3. Explain why a simple travel time does not predict damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A converging lens has f=15 cm and d_o=6 cm. Use $1/f=1/d_o+1/d_i$ and $m=-d_i/d_o$. Find d_i, magnification and image kind.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: d = vΔt; Δt = d/v.
Use the supplied wave speed for its medium and wave type.
These classroom models are not real hazard forecasts.

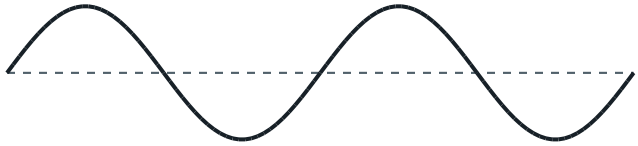
Worked reasoning: In a simple model, a wave amplitude falls from 4 to 2 units while intensity is proportional to amplitude squared. Intensity becomes one quarter, not one half.

Your turn: Draw the two waves and two corresponding relative-intensity bars.

Marzano check - circle: ① ② ③

03 Represent and interpret

Draw a disturbance traveling through water and mark local water motion. Why does wave arrival not mean the original water crossed the whole ocean?



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Solve and justify

Supplied seismic-speeds: P=6 km/s, S=3.5 km/s. At a station 210 km away, find arrival times and S-P delay.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

Why does knowing wave arrival time alone not establish local damage? Name other evidence.

Marzano check - circle: 1 2 3

06 Choose a method

Light moves from a transparent medium with n=1.7 into air (n=1). Find critical angle using $\sin\theta_c = 1/n$. Describe incidence above it in the ideal model.

Marzano check - circle: 1 2 3

07 Exit check

A shallow-water model uses $v=\sqrt{(gh)}$. With g=10 and h=100 m, find speed and time for 10 km. State a limitation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A signal has intensity 8 W/m² at 10 m from an ideal point source. Find intensity at 30 m and total radiated power. Explain one limitation for ground waves.

Marzano check - circle: 1 2 3